

Basic Info

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: <https://hajekel.github.io/>

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Dutch robotics engineer with a mechanical engineering background. Specialized in teleimpedance-robotics, autonomous robots, control systems, motion planning, ROS, deep learning, and computer vision. Developed various [projects](#) integrating these fields.

I am seeking a role as a robotics engineer within a dynamic and innovative company that offers opportunities to work alongside experienced robotics engineers from whom I can learn and grow. I am particularly interested in a company that supports professional development through attendance at industry conferences and provides opportunities for obtaining further certifications or diplomas, funded by the company. More information is available at my portfolio website: <https://hajekel.github.io>.

Experience

MAR 2025– NOW	Robotics Software Engineer <i>Demcon Unmanned Systems, Scheveningen, Netherlands</i> Developed code for the V2000 , V3000 , V5750 , V9975 USV's.
SEP 2023–SEP 2024	Robotics Engineer <i>IamMirte, Delft, Netherlands</i> Engineered the MIRTE master robot's software and hardware, enhancing the practical learning experience for 120 robotics students. This role demanded a continuous growth mindset, integrating new technologies such as Docker, X11 forwarding, UART, PCB assembly, Servo Motor ID Programming, URDF generation, and mecanum wheel control implementation. Hard Skills : Linux, Git, ROS, C++, SSH, X11 forwarding, design, robot dynamics and control.
SEP 2023–Nov 2023	AI System Developer <i>KLM, Amsterdam, Netherlands</i> Developed and implemented an adaptive AI interactive voice response system , aiming to reduce KLM's employee training costs by 30%. Demonstrated critical thinking through the design of a minimal working example that met customer requirements and continuously improved it for real-time, human-like interaction. Utilized creativity in prompt engineering, enabling the telephone bot to simulate various customer scenarios using disfluencies for more realistic behavior. Demonstrated problem-solving skills by creating the first IVR using VOIP for a real-time, human-like call bot based on generative AI. Collaborated with a diverse team from India, Wales, and Iran, and a Dutch customer, requiring strong cultural awareness to integrate various perspectives effectively.  Hard Skills : Javascript, FastAPI, Python, Twilio, Streaming, HTTP, AWS, OpenAI API, VOIP, IVR, Agile Scrum.
SEP 2023–Nov 2023	MSc Robotics Teaching Assistant <i>Delft University, Delft, Netherlands</i> Facilitated courses in Robot Dynamics and Control, Deep Learning, and Multi-disciplinary Projects for over 300 students. This role required leadership to guide students through complex concepts and projects. For instance, set up meetings with a PhD researcher to clarify tasks and align solutions when students faced unclear assignments. Demonstrated emotional intelligence by assessing each student's understanding and using tailored analogies to explain abstract concepts. Worked with students from 21 different nationalities, necessitating strong cultural awareness to effectively communicate and empathize with diverse backgrounds and learning styles. Hard Skills : Python, systems and control, arm/vehicle kinematics and dynamics, advanced manipulation.
APR 2023–JUL 2023	Robotics Engineer <i>Ahold Delhaize, Delft, Netherlands</i> Developed a comprehensive order-picking and assistance system for the Albert robot in supermarkets as the Project Manager and Computer Vision Specialist, demonstrating strong leadership through the coordination and direction of the project. The robot handles both online and in-store customer orders, utilizing advanced voice interaction powered by ChatGPT for in-store requests and a user interface for online orders. It autonomously identifies, picks, and places products with the FlexBE system, returning to a base station after completing orders. Equipped with lidar, radar, and stereo cameras, the robot detects customers and obstacles. Effective communication skills were essential for collaboration with team members and stakeholders. Testing in simulations and real-world scenarios proved the system's reliability in enhancing the robot's capabilities, highlighting problem-solving skills when troubleshooting technical issues and optimizing system performance. Additionally, attention to detail was crucial in ensuring the robot's precise operation and reliability in diverse environments. Hard Skills : ROS, C++, Python, Deep Learning, Computer Vision, Sensor Fusion, Decision-Making Algorithms.
AUG 2021–Nov 2021	AI Developer <i>Plaex, Enschede, Netherlands</i> Developed AI algorithms to enhance the accuracy and efficiency of Garby, an automatic waste sorting system. Hard Skills : Deep learning, YOLOv4, Mechanical Design, Computer Vision.

SEP 2019–JUL 2021	BSc Mechanical Engineering Teaching Assistant <i>University of Twente, Enschede, Netherlands</i> Provided instructional support in BSc mechanical engineering, improving student comprehension and application of mechanical principles through teaching methods and practical demonstrations.
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Education

2021–2024	MSc. in Robotics <i>Delft University, Delft, Netherlands</i> Specializations in deep learning, computer vision, autonomous flight of micro air vehicles, tele-robotics and haptics, knowledge representation, and symbolic reasoning. Experience in motion planning, human-robot interaction, control theory, ethics, and multidisciplinary teamwork. Hard skills: C++, Python, ROS, Machine Perception, Autonomous Flight of Micro Air Vehicles, Deep Learning, Computer Vision, Machine Learning, Localization, Bayesian state estimation, path planning, trajectory design, sensor fusion. Thesis: <i>Visio-Verbal Teleimpedance Interface: Enabling Semi-Autonomous Control of Physical Interaction via Eye Tracking and Speech</i> . Published in Frontiers in Robotics and AI and on arXiv . Collaboration between Delft University and CERN.  Literature review:  . Published Paper:  . Extended paper:  . Supervisor: Luka Peternel . Grade: 8
2021	Deep Learning Specialization <i>Online, Coursera</i> Completed the Deep Learning Specialization on Coursera, offered by DeepLearning.AI and led by Andrew Ng. Acquired comprehensive knowledge of deep learning, covering Convolutional Neural Networks, Recurrent Neural Networks, LSTMs, Transformers, and advanced techniques like Dropout and BatchNorm. Gained practical experience in building and training neural network architectures, implementing them with Python and TensorFlow, and applying these models to tasks such as speech recognition, music synthesis, and natural language processing. Equipped with the skills to develop and optimize deep learning applications effectively. Hard Skills : CNNs, RNNs, LSTMs, Transformers, Python, TensorFlow, NLP, Optimization Techniques, C++
2018–2021	BSc in Mechanical Engineering <i>University of Twente, Enschede, Netherlands</i> Specializations in Bio-robotics and Aircraft Engineering. Each quarter included a project related to the course subjects, where consistently took on the role of project leader. This experience significantly developed strong leadership qualities throughout the bachelor program. Hard Skills : Advanced calculus and differential equations, mechanics (statics and dynamics), thermodynamics, fluid mechanics, proficiency in CAD software (e.g., SolidWorks, AutoCAD), 3D modeling and simulation, programming (e.g., MATLAB, Python), finite element analysis (FEA), computational fluid dynamics (CFD), structural analysis, kinematics and dynamics of machines, knowledge of manufacturing processes, production planning and quality control, control theory and systems, project planning and management, design methodologies and product development. Thesis: <i>Development and evaluation of visio-verbal teleimpedance interface using eye-tracking and LLMs</i> .  Supervisor: Bojana Rosic . Grade: 8

Languages

HUMAN	Dutch (<i>fluent</i>), English (<i>fluent</i>), French (<i>basic</i>), German (<i>basic</i>).
MACHINE	Python (PyTorch, Tensorflow, NumPy), C++, JavaScript, Matlab/GNU Octave, bash/shell, $\LaTeX$, R Markdown, HTML, CSS.

Awards

2023	Awarded first prize for best AIRLab robot design in MSc robotics. <i>Ahold Delheize Delft, Netherlands</i> First prize for best AIRLab robot design of the MSc robotics.
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- Technologies: ROS, OpenAI, ComputerVision.

Familiar Tools

Usual Workflow

Very familiar with **Docker** containerization and orchestrating them using **Docker Compose**. Relies on **Markdown** and **L^AT_EX** for documentation. For coding, uses **VSCode** as a code editor and **Git** version control. Uses coding languages **MATLAB**, **C++** and **python** for robotic backend applications and **javascript** **HTML**, **CSS**, **JavaScript**, **JSX**, and **TypeScript** for frontend applications. Experienced across various operating systems but uses **GNU/Linux** on a daily basis. Regularly uses the **SSH** network communication protocol to communicate with other computers. Experienced with **ROS** and manages sessions with a terminal multiplexer (**Tmux**).

Programs I am Familiar With

Proficient in **SolidWorks** for computer aided design (CAD) and experience with **Blender** for creating complex renders, such as an [aircraft model](#). Managed websites through tools like **GitHub Pages**. Experience in training neural networks via google compute engine. Familiar with **ROS** for robotics middleware, **Simulink** for simulation and model-based design, and tools for **Deep Learning and Machine Learning** including **PyTorch** and **TensorFlow**.

Interests

Passionate about daily sports and a healthy lifestyle. Enjoys making music, gardening, and Latin dancing.